

RESEARCH ARTICLE

Seizure monitoring by combined diary and wearable data: A multicenter, longitudinal, observational study

Solveig Vieluf^{1,2,3}  | Sasagu Tomioka⁴  | Bo Zhang⁵ | Vaishnav Krishnan⁶ | William J. Bosl^{7,8} | Todd Grinnell⁴ | Tobias Loddenkemper¹

¹Division of Epilepsy and Clinical Neurophysiology, Boston Children's Hospital, Harvard Medical School, Boston, Massachusetts, USA

²Department of Medicine I, Ludwig Maximilian University Hospital, Ludwig Maximilian University Munich, Munich, Germany

³DZHK (German Centre for Cardiovascular Research), partner site, Munich Heart Alliance, Munich, Germany

⁴Sumitomo Pharma America, Inc., Marlborough, Massachusetts, USA

⁵Department of Neurology, Boston Children's Hospital, Harvard Medical School, Boston, Massachusetts, USA

⁶Departments of Neurology, Neuroscience, and Psychiatry & Behavioral Sciences, Baylor College of Medicine, Houston, Texas, USA

⁷Data Institute, Clinical Neuroinformatics & AI Laboratory, University of San Francisco, San Francisco, California, USA

⁸Computational Health Informatics Program, Boston Children's Hospital, Harvard Medical School, Boston, Massachusetts, USA

Correspondence

Sasagu Tomioka, Sumitomo Pharma America, Inc. 84 Waterford Drive, Marlborough, MA 01752, USA.
Email: sam.tomioka@us.sumitomo-pharma.com

Funding information

Sumitomo Pharma America, Inc. (formerly Sunovion Pharmaceuticals Inc.); American Epilepsy Society, Grant/Award Number: 932267; Epilepsy Research Fund

Abstract

Objective: In patients with intractable epilepsy, accurate diaries of seizure occurrence and timing can substantially inform management. Wearable devices that provide confirmation of seizure occurrence complement such diaries, which are frequently incomplete and/or inaccurate. Here, we combined seizure diaries and longitudinally deployed wrist-worn device recordings to evaluate whether wearable recordings contain information that can discriminate between days containing seizure-related activity and those without.

Methods: Patients with focal seizures were prospectively enrolled in a clinical trial to test the effectiveness of eslicarbazepine acetate as an adjunct to levetiracetam or lamotrigine (phase IV clinical trial NCT03116828). One hundred two patients maintained a seizure diary and wore a biosensor for >30 weeks. Based on diaries, we labeled days as either “seizure” versus “no-seizure” or “preseizure” versus “no-preseizure.” We compared patterns obtained by harmonic 24-h modeling between conditions. Best-ranking wearable markers and seizure diary variables were fed into a fully connected neural network, with several hidden layers and depth as hyperparameters that classified between seizure day conditions.

Results: The final sample contained 70 patients (median age = 42.5 years, 43 female) with 5437 recorded patient-days, including 557 seizure days and 537 preseizure days. Twenty-four-hour patterns in electrodermal activity and accelerometry

Solveig Vieluf and Sasagu Tomioka contributed equally to this work.

Todd Grinnell and Tobias Loddenkemper jointly supervised this work.

© 2025 International League Against Epilepsy.

differentiated no-seizure versus seizure days, as well as no-preseizure versus preseizure days ($p < .01$). Classification between no-seizure and seizure days (weighted F1 = .81, sensitivity = .82, specificity = .67) as well as between no-preseizure and preseizure days (weighted F1 = .82, sensitivity = .80, specificity = .66) revealed good performance.

Significance: Wearable data capture seizure-related differences with daily resolution, differentiating between days with lower and higher seizure susceptibility. Combining diary-based clinical and wearable data bears the potential for developing a dynamic seizure detection and prediction system with daily resolution.

KEYWORDS

24-h pattern, biosensors, eslicarbazepine acetate, seizure biomarker, seizure forecasting

1 | INTRODUCTION

Epilepsy is a neurological disorder that affects millions of people worldwide, one third of whom experience persistent seizures despite antiseizure medications (ASMs). Accurately tracking seizure occurrence plays an essential role in rigorously validating novel therapies and in the management and selection of ASMs.¹ To this date, seizure diaries provide the main outcome measure for epilepsy, but diaries lack accuracy, particularly over extended periods.² Both underreporting and overreporting complicate the interpretation of diary information.³

Emerging technologies aim to provide improved epilepsy management. More reliable avenues for seizure monitoring related to chronic intracranial or subscalp electroencephalographic (EEG) recordings are available.^{4–6} However, these invasive strategies are not without risk and often fail to discriminate between electroclinical and pure electrographic events. Wearable technologies have surfaced as an innovative and less invasive potential solution to complement traditional seizure diaries.^{7,8} Wearables can capture seizure-related changes in autonomic and actigraphic data before and after seizures and enable seizure detection and prediction, respectively.^{9–12} Although prior studies demonstrate feasibility in inpatient settings using EEG as the gold standard, challenges remain in ambulatory or real-life settings where the best data labeling comes from diaries, with a daily resolution and potential limitations of self-reported seizures. Recent research has shown that seizure occurrence is driven by cycles of different period lengths in diaries, EEG, and wearable, providing new avenues for improving seizure tracking and likelihood calculations.^{13–16}

Our previous work identified potential biomarkers from inpatient recordings that may enhance seizure

Key points

- The analysis includes a robust dataset spanning over 5000 days from a retrospective observational cohort study.
- Seizure monitoring in outpatient settings can be enhanced by integrating wearable data with retrospective diary entries, as this approach offers mutual benefits.
- Distinct patterns in seizure days before another seizure day suggest the potential for defining a novel seizure day category.

detection and prediction strategies. First, we modeled 24-h patterns to capture circadian rhythms and showed that these patterns can distinguish between seizure and nonseizure recordings.^{17,18} Second, ultradian rhythms, with periods between 1 and 12 h, differ between recordings with and without seizures in specific frequencies.¹⁹ Third, we discovered that short evening recordings could forecast whether a seizure was likely to happen the next day.²⁰

Here, we aimed to test these three markers' potential for seizure detection and prediction in a longitudinal real-life outpatient dataset. We sought to determine whether no-seizure days differed from seizure or preseizure days regarding 24-h patterns, ultradian characteristics, and short evening recordings of wearable data (aim 1). To test the potential of the markers showing differences between seizure conditions for seizure detection and prediction, we aimed to classify between no-seizure and seizure or preseizure days, respectively (aim 2).

2 | MATERIALS AND METHODS

2.1 | Study design

This is a retrospective, observational cohort study. Data were collected during a 31-week, multicenter (39 sites in the United States and one site in Canada), two-arm, prospective, open-label, nonrandomized, phase 4 study in the United States and Canada (phase IV clinical trial NCT03116828). Details on the study design can be found online (see data sharing section). Patients were enrolled in two study arms. The underlying trial aimed to test the effectiveness of eslicarbazepine acetate (ESL) as adjunctive therapy to levetiracetam or lamotrigine in adults with focal seizures over a 24-week maintenance period to replicate a traditional phase 3 study arm. To maximize the dataset, we merged both trial arms, as the monitoring design, including wearable and diary, were similar for both arms, and evaluated the dataset as a single cohort. For this trial, wearable data were purely exploratory endpoints and were not envisioned to influence the results or the interpretation of primary endpoints. We adhered to the STROBE checklist.

2.2 | Patients

We included ESL-naïve patients with intractable focal epilepsy (aged 18 and older) with a documented EEG supporting the diagnosis, within the past 10 years and at least three focal seizures during the preceding 6 months. At the time (2016) of the phase 4 open-label clinical trial design from which these data were extracted, the most recent version of the 2017 International League Against Epilepsy (ILAE) Classification of Seizures had not yet been fully implemented. The terminology used in the protocol development reflected the existing accepted nomenclature from the 2010 update to the 1981 ILAE classification system. This was also the case with all previous clinical trials of ESL. It was also consistent with the version of the classification system used in the approved US Food and Drug Administration indication for the use of ESL. Because seizures were identified and classified in this trial using the older terminology, it would not be appropriate to assign the newer terminology retrospectively. Neuromodulatory devices (e.g., vagus nerve stimulator) had to be implanted more than 6 months previously. Patients were excluded if they had taken a medication that was prohibited in the protocol of the clinical trial if they had seizures of generalized onset, exclusively focal seizures without motor signs, a history of status epilepticus or cluster seizures, or nonepileptic seizures. In addition, patients with other progressive physical or mental disorders were excluded. See Figure 1 for inclusion tree and dataset overview.

2.3 | Procedures

Data were recorded and collected from 2017 to 2019. For all included patients, age, sex, epilepsy duration, seizure frequency, etiology, and worst seizure type were collected based on medical records and patient self-report. Additionally, medication intake history and seizure diaries were obtained. The seizure diary included the seizure date, estimated time, seizure type, and additional notes. Patients received paper seizure diaries during screening, titration, and every 4 weeks during the 24-week maintenance phase. We retrospectively calculated prestudy seizure frequency from historical patient diaries. Diaries were collected and analyzed on the day of the next visit. Patients indicated whether and how many seizures they had with daily accuracy. Patients alternately wore two Embrace (Empatica) sensors that recorded electrodermal activity (EDA; sampling rate of approximately 4 Hz), wrist temperature (TEMP; sampling rate of approximately 1 Hz), and three-axis accelerometry (sampled at 32 Hz). During enrollment, defective devices were replaced. Wearable data were analyzed with MATLAB 2023b (MathWorks). For each patient, we concatenated all data. In overlapping periods, we retained the recording with the higher temperature, assuming the wrist temperature exceeded room temperature. We removed data points in all three signals if the TEMP was <24 or $>37^{\circ}\text{C}$, and we removed EDA data $<.01$ or $>40\ \mu\text{S}$.^{17,21} We included days with >8 h of quality recording that could be discontinuous. We converted continuous three-dimensional acceleration data into a single acceleration vector (ACC) by calculating the root sum of squares.¹⁹ Based on diary entries, we divided time series data into 24-h windows (midnight to midnight), annotating them as no-seizure versus seizure and no-preseizure versus preseizure day. Preseizure and seizure days could be the same and we labeled them as “both” for follow-up analysis. See Figure 2 for a methods illustration.

2.4 | Wearable data analysis, statistics, and machine learning

For group-based comparisons of 24-h patterns between no-seizure and preseizure and seizure days, respectively, we took the median of each 10-min window. We modeled 24-h patterns with harmonic models, with two harmonic terms for EDA and one harmonic term for ACC and TEMP per patient, and saved model parameters.¹⁷ To compare the seizure-day conditions among each other, we fit a nonlinear mixed-effects harmonic model to either all days or a model that included a group, that is, seizure-day condition indicator. We decided on the better fitting model as determined by the rejection decision parameter h and the p -value.¹⁷

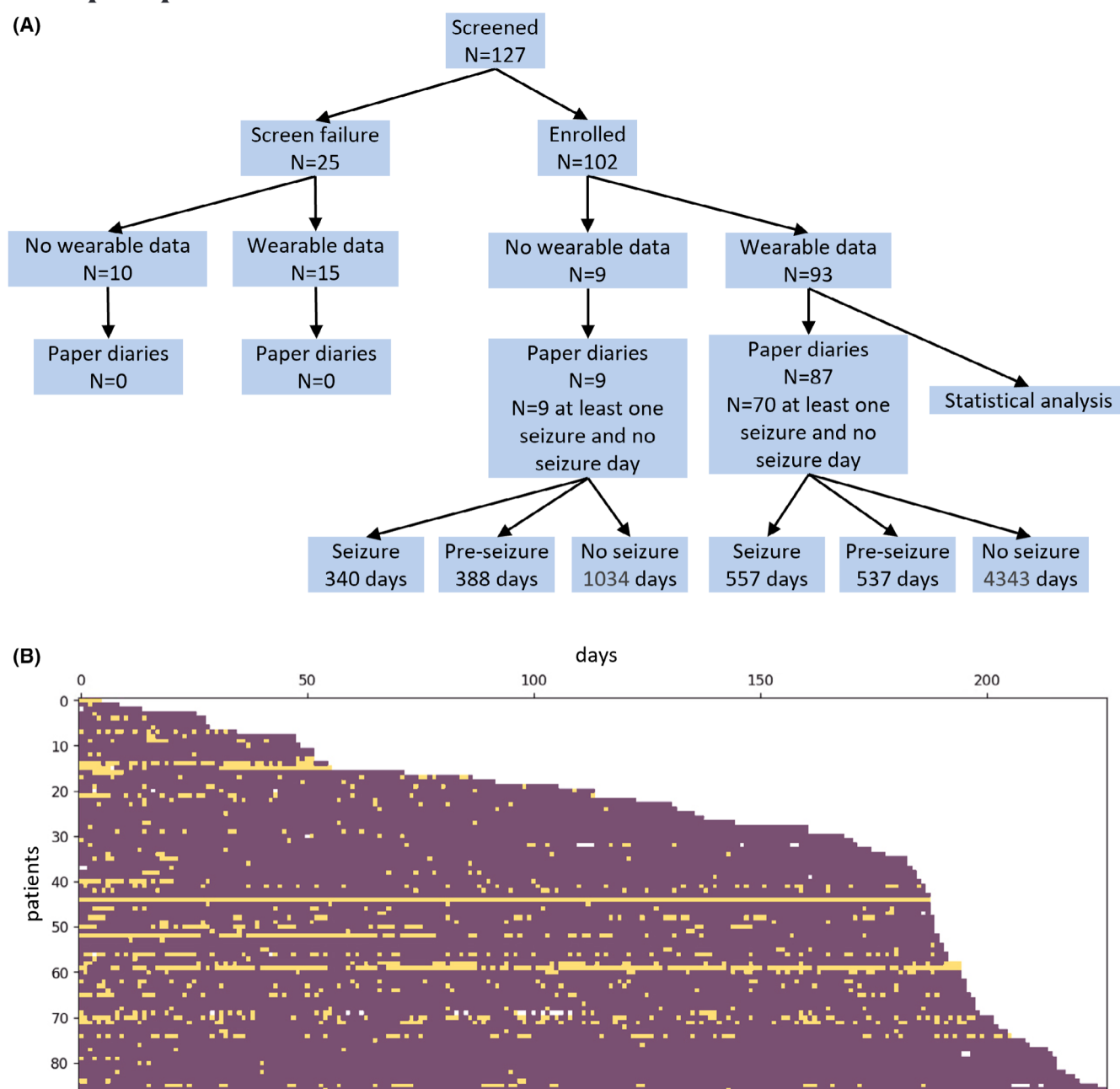


FIGURE 1 Dataset overview with (A) inclusion diagram and (B) dataset overview showing the number of days covering the period of diary entries by the patient (yellow indicates a seizure day, and purple represents a no-seizure day), before excluding days based on wearable data availability and quality.

We calculated the power spectral density to reveal ultradian patterns by plotting Lomb–Scargle periodograms with period lengths from .125h to 12h in .125-h increments. We extracted peak power and peak period length around 12-, 10-, 4-, 3-, and 2-h period lengths with a window of $\pm .25$ h, when a difference was visually detected. Power values were compared with the paired two-sample *t*-test and period length with the Wilcoxon signed rank test. For short recordings, in addition to a 10-min time window at 9p.m. as previously reported,²⁰ we selected a time window at 11:30p.m. as the average adult bedtime.

A paired two-sample *t*-test was performed to compare pre-seizure and no-seizure day recordings. For reporting and inclusion into the next step of deep neural network training, we set the significance level to $p < .05$.

We used Python with Google TensorFlow version 2.8.2 and Keras²² with Keras-tuner,²³ as well as Scikit-learn²⁴ to run a deep neural network algorithm for the classifications. We included wearable markers that were significant on a group level combined with demographics, clinical data, and the diary-based days since the last seizure variable. Seizure detection and prediction were

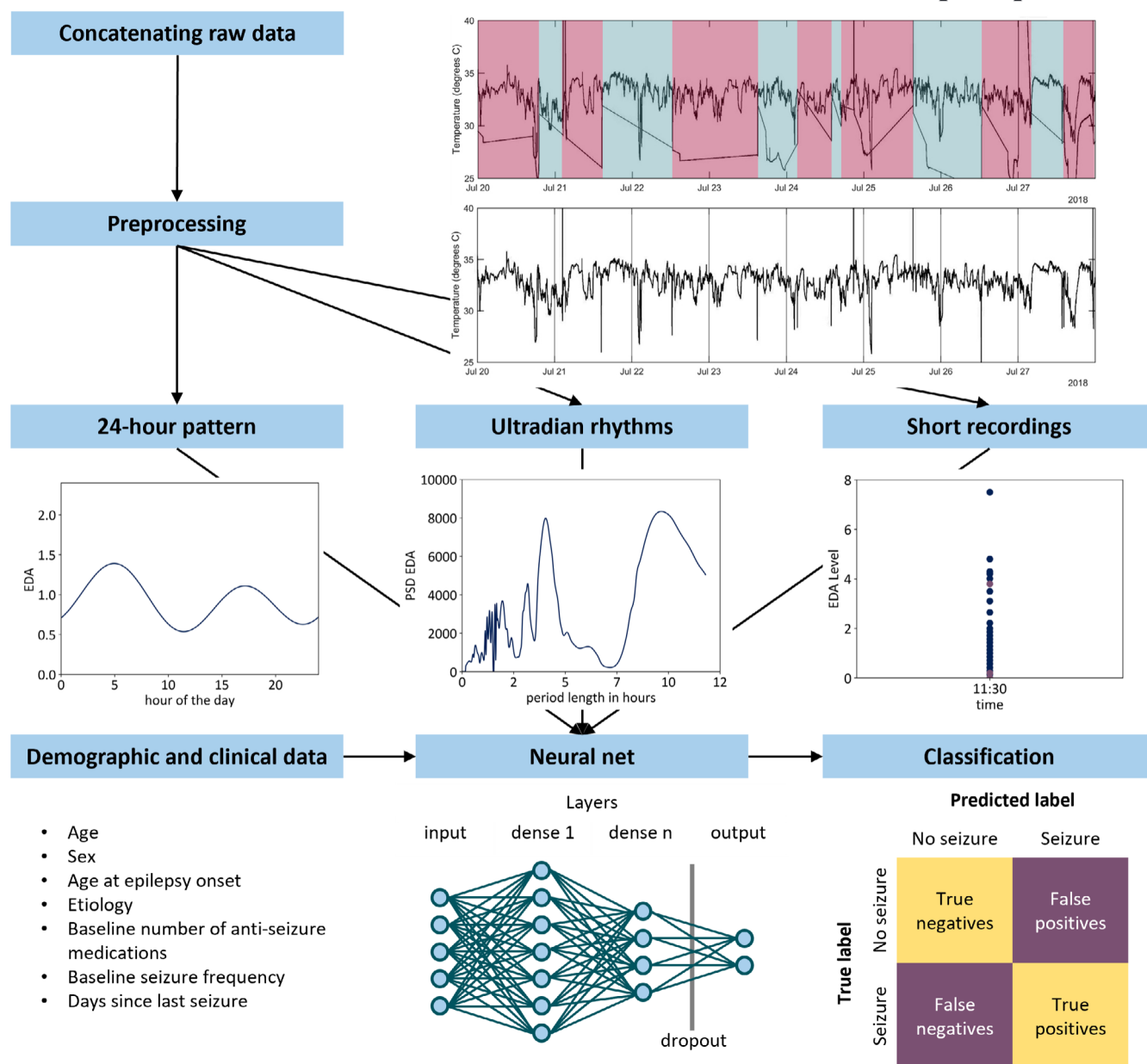


FIGURE 2 Summary of the analysis pipeline, including data preparation and preprocessing, before biomarker validation by analyzing 24-h patterns, ultradian rhythms, and evening short recordings. Biomarkers showing significant differences between seizure-day conditions (no-seizure and pre-seizure or seizure) are used as input features combined with demographic and clinical data for classifications between seizure-day conditions by a fully connected neural net. EDA, electrodermal activity. PSD, pre-seizure day.

based solely on retrospective seizure diary entries, using only information available up to the previous day, without incorporating data from the day being evaluated. We excluded etiology and worst seizure type due to subgroups of less than three patients and the EDA short recording, as it highly correlated ($p < .01$) with the EDA 24-h level. `sklearn.feature_selection.RFECV` performs recursive feature elimination with cross-validation, systematically removing less important features and evaluating model performance at each step to select the optimal subset of features that maximizes predictive accuracy. We performed `sklearn.feature_selection.`

`RFECV` on randomly oversampled data with 10 runs and included a minimum of six features for each fold for detection, prediction, and seizure activity type classification separately. The feature selection was performed for combined data and wearable data analysis only. See [Figure S1](#) for an illustration of the machine learning pipeline.

For network training, we used random oversampling. We performed stratified fivefold cross-validation, splitting the dataset into 80% training data and 20% test data. The training data was further split (also stratified) into 90% actual training data and 10% validation data for

the fit. We chose a per-day split approach for the general model, which allows data from the same patient to be included in the training, validation, and test sets. Despite the potential risk of data leakage, this approach was selected because the number of days per patient varied significantly, making stratification impractical given the 9:1 split in the inner loop and the 4:1 split in the outer loop. Additionally, an entire patient was held out for independent testing in the leave-one-out approach. Although the per-day split does carry a risk of minor information leakage due to shared patient data across subsets, the observed robustness of the model's performance suggests that it maximizes data usage and effectively captures general patterns rather than overfitting to patient-specific data. The similarity of the results between the per-day and leave-one-out approaches would indicate that interpatient-specific data were not critical for model performance.

We trained fully connected deep neural networks for different objectives. Each neural net consisted of N dense layers of width W , an optional dropout layer with an adjustable dropout rate, and a dense output layer using the softmax activation function. We used the sigmoid activation function for the N hidden layers. We used the Adam optimizer with a categorical cross-entropy loss function. The number of layers, their width, the use of the dropout layer, and its dropout rate, as well as the batch size and the learning rate, were tunable parameters in this study. We used `keras_tuner` and the `BayesianOptimizer` to perform a hyperparameter scan based on a probabilistic model, optimizing for the number of “true positives.” We used Keras' early stopping function, with a “patience” of 50 epochs, to terminate training when the validation loss reached its minimum. The performance metrics were evaluated on the test data for each fold and average. We report weighted F1, weighted precision, weighted recall, and the area under the precision–recall curve (AUC_PR), as well as sensitivity and specificity as model performance parameters. For seizure activity comparisons, we report weighted F1, weighted precision, and weighted recall. For model evaluation, we chose the weighted F1 score as an overall performance metric and decided to consider weighted F1 of $<.8$ as poor performance, $.8$ up to $.9$ as good, and $>.9$ as excellent.

3 | RESULTS

3.1 | Patients and recordings

One hundred twenty-seven patients were screened, and 102 subjects were enrolled in the study. Of those, wearable data and/or paper diaries were missing for 15

patients (for more details, see [Figure 1](#)). Seventy-seven patients had at least one no-seizure and one seizure day recorded and were screened for data quality. The final sample comprised 70 patients, with a total of 5437 recorded days (out of an original 12137 patient-days within the reporting period, resulting in a deficiency time of 55%). Among these, 557 were seizure days, 537 were pre-seizure days, and 242 were classified as both seizure and pre-seizure days ([Figure 1](#)). The average age of the included patients was 42.5 years, and 43 were female (for demographics and clinical characteristics, see [Table 1](#)).

3.2 | Daily autonomic nervous system activity differences between no-seizure and seizure or pre-seizure days

The analysis of 24-h patterns revealed that the model differentiating between groups (H1) was superior for EDA ($h=1$, $p<.01$) and ACC ($h=1$, $p<.01$), whereas H0 represents data better for TEMP ($h=0$, $p=1$; [Figure 3](#)). For ultradian rhythm analysis, the overall power did not differ between seizure conditions (all $p>.05$). We observed lower power at seizure than no-seizure days for EDA 4-h and ACC 10-h cycles. Peak period lengths differed for EDA 12- and 4-h, ACC 4- and 2-h, and TEMP 12- and 3-h cycles. Pre-seizure days had lower ACC power for 10-h, and peak period lengths differed for EDA 12- and 4-h, ACC 4-h, and TEMP 12- and 3-h cycles ([Table S1](#)). Comparing pre-seizure and no-seizure evening recordings did not differ at 9 p.m. ($t=1.08$, $p=.28$), but EDA levels were higher in pre-seizure compared to no-seizure nights at 11:30 p.m. ($t=2.29$, $p=.03$).

3.3 | Classifications for seizure detection and prediction

See [Table S2](#) for feature ranking; all features with rank 1 are included in the fold analysis. Features selected over folds always contained days since last seizure and baseline seizure frequency as well as parameters of the 24-h EDA and ACC patterns. Performance characteristics of the machine learning are summarized in [Table 2](#). For the classification between no-seizure and seizure days relevant for seizure detection, the classification performance was comparable for the combined feature input (AUC_PC = .84) and stochastic diary-based input (AUC_PC = .83), and did not allow distinguishing between seizure and nonseizure days based on wearable data only (AUC_PC = .49).

TABLE 1 Demographic and clinical characteristics of 102 enrolled patients and 70 patients with wearable data and paper diaries included in the analysis.

		Enrolled, <i>n</i> = 102	Final analysis, <i>n</i> = 70
Age, years	Median	40.5	42.5
Sex, <i>n</i> (%)	Female	60 (58.8%)	43 (61.4%)
	Male	42 (41.2%)	27 (38.6%)
Epilepsy duration, years	Mean (SD)	15.1 (14.85)	17.1 (14.81)
<i>n</i> (%)	≥20 years	29 (28.4%)	24 (34.3%)
	<20 years	73 (71.6%)	46 (65.7%)
Etiology, <i>n</i> (%)	Hereditary	6 (5.9%)	2 (2.9%)
	Trauma	21 (20.6%)	16 (22.9%)
	Infection	2 (2.0%)	1 (1.4%)
	Brain tumor	2 (2.0%)	1 (1.4%)
	Temporal sclerosis	1 (1.0%)	1 (1.4%)
	Febrile seizure	2 (2.0%)	0 (.0%)
	Cerebral	6 (5.9%)	3 (4.3%)
	Brain malformation	7 (6.9%)	5 (7.1%)
	Unknown	56 (54.9%)	40 (57.1%)
	Other	4 (3.9%)	4 (5.7%)
Worst seizure type [baseline], <i>n</i> (%)	Simple partial	10 (9.8%)	8 (11.4%)
	Complex partial	66 (64.7%)	47 (67.1%)
	Partial evolving to secondary generalized	24 (23.5%)	14.0 (20.0%)
	Unknown	1 (1.0%)	0 (.0%)
	Other	1 (1.0%)	1 (1.4%)
Baseline standardized seizure frequency [28 days]	Mean (SD)	11.9 (41.79)	10.2 (40.41)
	Range	.3–337.5	.3–337.5
Number of documented seizure days	Mean (SD)	19.6 (32.71)	17.9 (31.77)
	Range	1–188	1–188
AED use, count	Mean (SD)	1.4 (.60)	1.5 (.63)

Note: This table reflects the data capture terminology used at the time of the trial.

Abbreviation: AED, antiepileptic drug.

For the classification between no-seizure and pre-seizure days, the performance was similar for combined wearable and clinical input data (AUC_PC=.87), and for diary-based input, that is, the stochastic model (AUC_PC=.83), and did not classify based on wearable data only (AUC_PC=.68).

3.4 | Differentiability of seizure activity types

As a follow-up analysis, we compared pre-seizure days, seizure days, and days that are both pre-seizure and seizure days. The classification performance was comparable for combined wearable (AUC_PC=.48) and clinical input and diary-based input (AUC_PC=.51) and did not classify based on wearable data only (AUC_PC=.33).

Classification results are summarized in Table 2 and Figure 4 as well as Figure S2. Feature selection for wearable data only is presented in Table S2. Figure S3 and Table S2 show the results for the leave-one-out analysis that provided comparable results.

4 | DISCUSSION

We examined wearable data and confirmed differences in recordings captured during seizure or pre-seizure days compared to no-seizure days. Despite the group differences, the extracted wearable features lack discriminatory power for daily classification; the resolution of the diary data as the gold standard did not allow for finer granularity. Contextualizing wearable data with diary entries and clinical attributes significantly enhanced

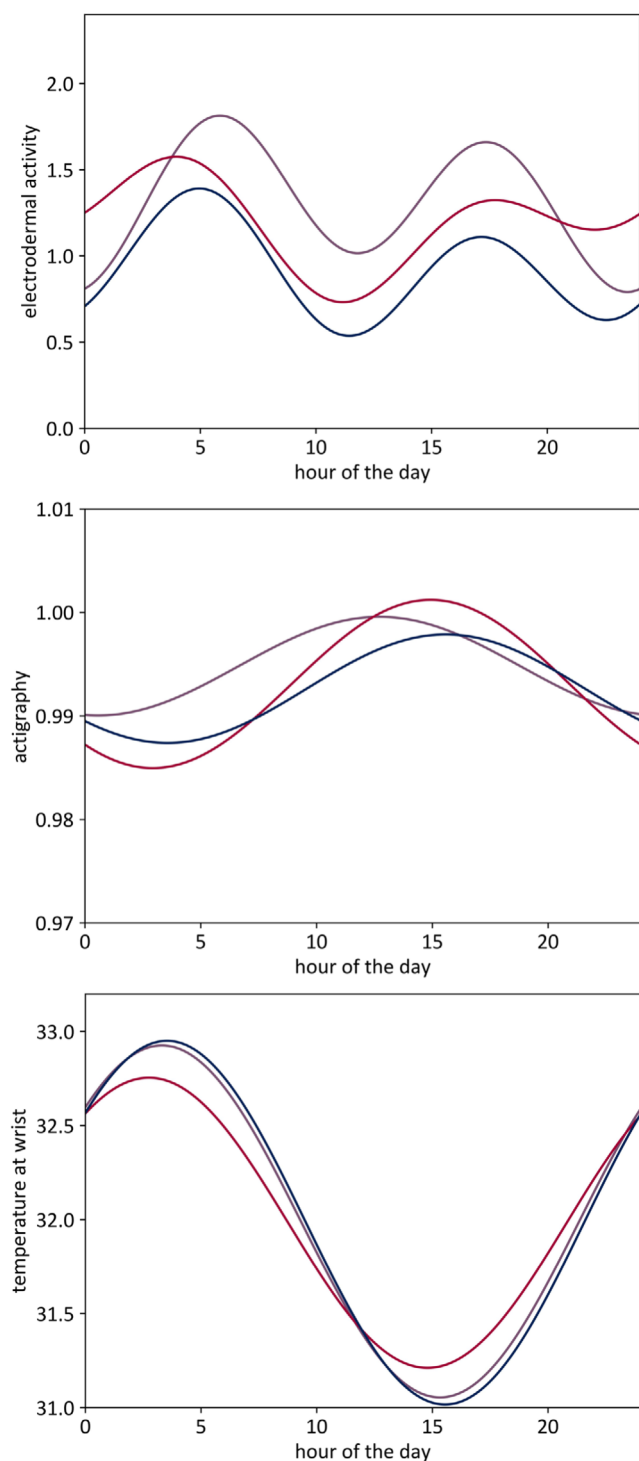


FIGURE 3 Results for modeled 24-h patterns (blue = no-seizure, purple = pre-seizure, red = seizure) for electrodermal activity, actigraphy (acceleration vector), and temperature at wrist data.

classification performance to a similar level as using diary-based features alone. Classification based on historical diary data, that is, days since last seizure and seizure frequency, allowed discrimination between seizure activity conditions and no-seizure days. Furthermore,

our analysis revealed distinctive patterns for combined seizure and pre-seizure days, hinting at potential additive or multiplying effects of consecutive seizures. These findings underscore the importance of integrating multimodal data sources for comprehensive seizure management strategies.

4.1 | Seizure diaries contain predictive information

Despite many shortcomings, seizure diaries are the standard of care in epilepsy management and build a valid source of information for clinical trial outcomes. Epilepsy is characterized by cyclic dynamics comprising phases with both high and low seizure likelihoods.^{25,26} Based on self-reported data from a mobile seizure diary, circadian patterns and multiday seizure patterns were extracted, and those contained personalized information on seizure likelihood.^{25,26} In line with this approach, we extracted seizure frequency determined at the start of the study and the daily changing variable of days since the last seizure from the diary data and built a classification system. In line with the cyclic approaches,^{13–16} we obtained good performance. We achieved a similar performance as a previously reported seizure diary alone approach that aimed at estimating the predictability of seizures by statistical and machine learning-based approaches.²⁷ Due to the imbalance in the dataset, we report weighted F1 and AUC_PR instead of the AUC_ROC, reported in other studies, for model performance, so a direct performance comparison is not feasible. The advantage of the approach presented in this study is that it can start from the first day of use of the system and it can learn and adjust easily over time. However, it would be of great interest to combine both approaches by adding the cycles after a recording of approximately 30 days. Here, we used paper-based diary data for labeling seizure days. However, some seizures may be missed, and reporting accuracy and reliability are low, as patients may not recall seizures, and some seizures may not be noted by caretakers.² The seizure diary used in this study included the seizure date, estimated time, seizure type, and additional notes. As the time of the seizures was estimated, the timing presented with inaccuracies that challenged the time-based seizure detection and prediction approaches.^{9,28} We attempted to overcome this challenge of the low temporal resolution of seizure diaries with day-based resolution biomarkers. For future studies, electronic seizure diaries offer the advantage of facilitating real-time data collection and analysis and add the importance of discriminating between seizure, no-seizure, and not-filled days.

TABLE 2 Performance metrics of machine learning models.

	Weighted F1		Weighted precision		Weighted recall		AUC_PR		Sensitivity		Specificity	
	Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI
Detection												
Stochastic	.83	.02	.89	.00	.80	.03	.83	.01	.82	.04	.67	.06
All ranked	.81	.02	.88	.01	.77	.03	.84	.02	.78	.03	.66	.06
Wearable	.59	.19	.82	.01	.56	.24	.49	.03	.57	.31	.44	.32
Prediction												
Stochastic	.83	.02	.89	.00	.79	.03	.83	.02	.81	.03	.67	.06
All ranked	.82	.02	.89	.00	.79	.04	.87	.02	.80	.04	.66	.06
Wearable	.72	.11	.82	.01	.69	.16	.68	.13	.73	.21	.29	.23
Seizure activity types												
Stochastic	.42	.04	.45	.04	.45	.03	.48	.04	N/A	N/A	N/A	N/A
All ranked	.42	.04	.44	.04	.44	.04	.51	.04	N/A	N/A	N/A	N/A
Wearable	.30	.04	.31	.06	.32	.03	.33	.02	N/A	N/A	N/A	N/A

Note: Means and 95% CIs are reported for weighted F1, precision, recall, AUC_PR, sensitivity, and specificity.

Abbreviations: AUC_PR, area under the precision–recall curve; CI, confidence interval; N/A, not applicable.

4.2 | Seizures disrupt the rhythmic daily activity of the autonomic nervous system

Various physiological processes in the human body are governed by complex interplays of different rhythms.²⁹ Human biological systems operate with precision through complex interplays of various rhythms, and polyrhythmicity is a distinguishing feature sensitive to events like seizures affecting internal balance. The potential seizure biomarkers, 24-h patterns, ultradian patterns, and short recordings were developed in a pediatric patient cohort who wore wearables during an inpatient stay at the epilepsy monitoring unit, and seizure patients had one or more tonic–clonic or focal impaired awareness seizures.^{17,18} In this study, wearable markers were tested on longitudinal outpatient data in an adult population with partial onset (focal) seizures. Here, we included peri-ictal data in the analysis, as the exact seizure times are unknown. We confirmed distinct 24-h patterns for electrodermal activity and showed patterns in actigraphy.^{17,18} The models' higher peaks for EDA and ACC on seizure days might reflect seizure activity. Ultradian patterns and short recordings differ between seizure activity and no-seizure days on a group level but did not contribute to the classification between conditions, which might relate to high intra- and interpatient variability.

Following moderate classifier performances for seizure detection and prediction and taking similarities of pre-seizure and seizure day polyrhythmic patterns into account, we tested the classification performance

between seizure and pre-seizure days. Many seizure days are followed by a seizure day, introducing a new seizure-related class (both). This indicates the high likelihood of acute repetitive seizures, that is, seizure patterns with periods of more frequent serial seizures.³⁰ Results showed a by-chance classification, indicating that days share common activity trends and that seizure activity builds up before the seizure happens. Additionally, the days that are both pre-seizure and seizure days have a higher likelihood of being deemed non-no-seizure days in the four-class comparison. This suggests additive or superimposed effects for our set of biomarkers and bears the potential to identify the lead-day of a period of more frequent serial seizures.

4.3 | Combining wearable with diary-based clinical data has the potential to make seizure monitoring more dynamic

Despite significant differences on a group level, the biomarkers we extracted from the wearable recordings did not differentiate between seizure-day conditions. Based on the diary as the gold standard, we can only classify with a daily resolution, making the results not directly comparable to previous seizure detection and prediction methods.^{4,5,12} Seizure diaries may permit the determination of seizure patterns that occur in circadian and multiday cycles, and seizure diary patterns may serve as predictors of seizure occurrence.^{8,25,26} Feature ranking shows that diary-based clinical data, that is, seizure

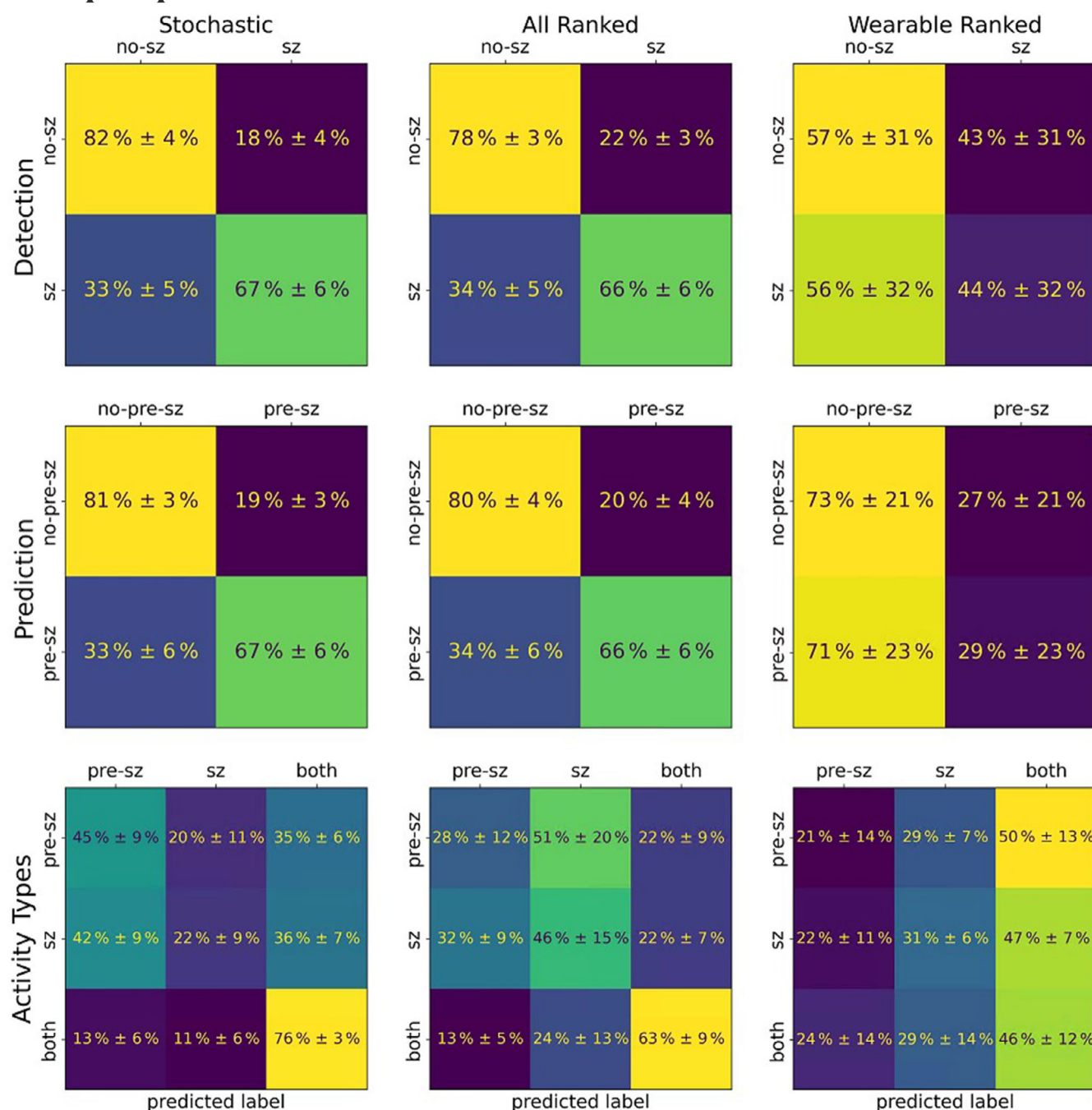


FIGURE 4 Summary of classification results by seizure activity-specific classifications for the feature combinations diary-based (stochastic), all features ranked, and wearable features ranked (from left to right). Validation results are averaged over five folds and normalized per row. sz, seizure.

frequency and days since the last seizure, are the two features with the highest importance. In line with the seizure cycle approaches, using a combination of patient-specific seizure frequency with the information of days since the last seizure derived from the historical seizure reporting revealed good accuracy. We conclude that seizure monitoring in outpatient settings can be enhanced by integrating wearable data with retrospective diary entries, as this approach offers mutual benefits; diary entries provide

context for wearable data, helping to identify patterns related to stress and other factors, and wearable data add real-time precision and flexibility to diary-based seizure tracking, capturing fluctuations in events and moods throughout the day. Our findings might serve as a basis for a prospective study, where multiday cycles based on wearable and diary data are added to the presented analysis.^{13,14,25} The prospective study could evaluate the correspondence between seizure-day labeling between diaries

and wearables. Incorporating a human-in-the-loop approach, where wearable devices prompt participants to report whether they experienced a seizure based on device predictions, could significantly enhance the accuracy of self-reported data. We believe that utilizing wearables to supplement traditional paper or app-based diaries could improve participant adherence and provide valuable real-time support for documentation. Future studies should explore this integration, as it holds promise for optimizing seizure monitoring and enhancing the quality of collected data. Also, future studies could explore the relationship between circadian patterns of seizures and the characterization of patients' chronotypes in relation to 24-h patterns in wearable data. This investigation would be of significant interest, as understanding these interdependencies could reveal how individual biological rhythms influence seizure occurrence, potentially leading to more personalized management strategies for epilepsy. Researchers may uncover critical temporal factors that impact seizure frequency and severity by integrating wearable data with chronobiological insights. The next step would be to test the clinical impact of an increased accuracy in seizure reporting. The clinical impact of improved seizure monitoring has not yet been tested in long-term prospective trials. Simulated data suggest that increased accuracy in seizure reporting does not directly affect ASM burden or seizure freedom rates³¹; however, the long-term impact may arise from better treatment adjustments enabled by higher accuracy, ultimately leading to improved outcomes. Beyond ASM burden, psychological factors such as quality of life and patient empowerment are critical aspects of clinical care, as enhanced seizure reporting accuracy may foster more informed patient–physician communication and enable more personalized treatment strategies.

4.4 | Limitations

Retrospective and partly exploratory analysis of wearable data collected during a clinical pharmaceutical trial presents several inherent limitations. First, the quality of the wearable data and patient compliance in wearing the device introduce noise and bias into the data. We excluded days with low-quality data, but we had no means to monitor compliance or how well the device was worn. The resolution of the diary was daily. Therefore, we restricted our analysis to seizure days due to the resolution of the seizure diary data. A follow-up study utilizing real-time seizure reporting, such as through an app, would enable more detailed, time-specific analysis. Additionally, we did not take other influencing factors, such as sleep, stress, or other temporal factors that were shown to impact seizure likelihood, into account.³² Using labels derived from

retrospective diary data can introduce recall bias, as participants may inaccurately report the timing or frequency of events like seizures, potentially skewing the results and weakening the predictive model's validity. This reliance on self-reported data may also miss subtle patterns or more minor seizures that participants do not notice. A future study could mitigate these issues by using objective, real-time data collection methods, such as continuous monitoring through wearable devices, implantable EEGs, or automated video analysis (at least for nighttime), providing an alternative and potentially more reliable gold standard for seizure detection. This approach would capture more precise, continuous data, allowing for a more accurate evaluation of predictive algorithms.

The retrospective nature of the analysis limits control and standardization of the conditions under which the data were collected. Also, the overall aim of the clinical trial was to test an early adjunctive therapy in adults with focal seizures. To maximize the dataset, we did not take the medication groups into account but used seizure frequency. Patients had comparably low residual baseline seizure frequency, and patients with generalized EEG-seizure onset were excluded. Both aspects might diminish the strength of the seizure-related signature in the wearable signal and increase the relevance of watch placement. Also, this selection bias impacts the generalizability to other patient populations. Here, most patients had multiple seizure subtypes, which increases the variability within patients. Due to variability in the dataset, we performed the feature selection as a preprocessing step based on the entire dataset, which incorporates information from the validation set during feature selection. Although this approach was chosen to ensure robust feature identification given the dataset's complexity and limited size, we acknowledge that it may impact the generalizability of the results and recommend that future studies validate the findings using independent test sets or cross-validation techniques if their dataset is larger and therefore the impact of individual patients on feature selection is minimized. All these aspects influence the generalizability of the models. One limitation of our study is the inability to compare the Embrace device to other seizure detection systems, as only retrospective data from this well-established and widely accepted device were available for analysis. To our knowledge, no established algorithms exist for seizure monitoring at the daily resolution we used, making direct comparisons with other approaches difficult. Furthermore, the wearable device did not record cardiac activity, which could greatly improve seizure monitoring.^{33,34}

Taken together, the retrospective analysis of wearable data offers valuable insights into real-world data; follow-up research must address various limitations to ensure the

validity and applicability of the findings, generated with this uniquely large dataset.

5 | CONCLUSIONS

Integrating diary-based, clinical, and wearable data shows differentiation between no-seizure and seizure/preseizure days, demonstrating the promising potential for improved seizure detection and prediction in the outpatient setting with the potential to contribute to better epilepsy management.

AUTHOR CONTRIBUTIONS

Solveig Vieluf, Sasagu Tomioka, Todd Grinnell, and Tobias Loddenkemper conceptualized and designed the study. Todd Grinnell and Sasagu Tomioka were involved in the data acquisition. All coauthors designed analysis methods used. Solveig Vieluf and Sasagu Tomioka performed data analysis. Solveig Vieluf wrote the initial manuscript draft. All coauthors drafted parts of the manuscript, edited the manuscript, and approved the final version.

ACKNOWLEDGMENTS

We thank Joanne Rahmati for her support.

FUNDING INFORMATION

The study was supported by the American Epilepsy Society under award number 932267, the Epilepsy Research Fund, and Sumitomo Pharma America, Inc. (formerly Sunovion Pharmaceuticals; phase IV clinical trial NCT03116828).

CONFLICT OF INTEREST STATEMENT

T.L. is part of patent applications to detect and predict clinical outcomes and to manage, diagnose, and treat neurological conditions, epilepsy, and seizures. T.L. has received past device donations from various companies, including Empatica, and has received research support from Empatica in the past. S.V., B.Z., and T.L. are part of a patent application covering technology for seizure forecasting. V.K. serves on the scientific advisory board of Enliten AI and is a paid consultant for the Digital In Vivo Alliance. S.T. is an employee of Sumitomo Pharma America, Inc. (formerly Sunovion Pharmaceuticals). Neither of the other authors has any conflict of interest to disclose. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

DATA AVAILABILITY STATEMENT

Deidentified data that support the findings of this study are available upon reasonable request, and approval pending intent of use. Study protocol is available at

<https://classic.clinicaltrials.gov/ct2/show/NCT03116828> and can be accessed by anyone for any purpose.

ORCID

Solveig Vieluf  <https://orcid.org/0000-0002-5532-8690>

Sasagu Tomioka  <https://orcid.org/0000-0001-6927-2096>

REFERENCES

1. Stirling RE, Maturana MI, Karoly PJ, Nurse ES, McCutcheon K, Grayden DB, et al. Seizure forecasting using a novel sub-scalp ultra-long term EEG monitoring system. *Front Neurol*. 2021;12:713794. <https://doi.org/10.3389/fneur.2021.713794>
2. Fisher RS, Blum DE, DiVentura B, Vannest J, Hixson JD, Moss R, et al. Seizure diaries for clinical research and practice: limitations and future prospects. *Epilepsy Behav*. 2012;24(3):304–10.
3. Hannon T, Fernandes KM, Wong V, Nurse ES, Cook MJ. Over- and underreporting of seizures: how big is the problem? *Epilepsia*. 2024;65(5):1406–14.
4. Kuhlmann L, Karoly P, Freestone DR, Brinkmann BH, Temko A, Barachant A, et al. Epilepsycosystem.Org: crowd-sourcing reproducible seizure prediction with long-term human intracranial EEG. *Brain*. 2018;141(9):2619–30.
5. Cook MJ, O'Brien TJ, Berkovic SF, Murphy M, Morokoff A, Fabinyi G, et al. Prediction of seizure likelihood with a long-term, implanted seizure advisory system in patients with drug-resistant epilepsy: a first-in-man study. *Lancet Neurol*. 2013;12(6):563–71.
6. Haneef Z, Yang K, Sheth SA, Aloor FZ, Aazhang B, Krishnan V, et al. Sub-scalp electroencephalography: a next-generation technique to study human neurophysiology. *Clin Neurophysiol*. 2022;141:77–87.
7. Stirling RE, Cook MJ, Grayden DB, Karoly PJ. Seizure forecasting and cyclic control of seizures. *Epilepsia*. 2021;62:S2–S14.
8. Brinkmann BH, Karoly PJ, Nurse ES, Dumanis SB, Nasser M, Viana PF, et al. Seizure diaries and forecasting with wearables: epilepsy monitoring outside the clinic. *Front Neurol*. 2021;12:690404.
9. Meisel C, El Atrache R, Jackson M, Schubach S, Ufongene C, Loddenkemper T. Machine learning from wristband sensor data for wearable, noninvasive seizure forecasting. *Epilepsia*. 2020;61(12):2653–66.
10. Tang J, El Atrache R, Yu S, Asif U, Jackson M, Roy S, et al. Seizure detection using wearable sensors and machine learning: setting a benchmark. *Epilepsia*. 2021;62(8):1807–19.
11. Vieluf S, Reinsberger C, El Atrache R, Jackson M, Schubach S, Ufongene C, et al. Autonomic nervous system changes detected with peripheral sensors in the setting of epileptic seizures. *Sci Rep*. 2020;10(1):1–8.
12. Poh M-Z, Loddenkemper T, Reinsberger C, Swenson NC, Goyal S, Sabtala MC, et al. Convulsive seizure detection using a wrist-worn electrodermal activity and accelerometry biosensor. *Epilepsia*. 2012;53(5):e93–e97.
13. Gregg NM, Attia TP, Nasser M, Joseph B, Karoly P, Cui J, et al. Seizure occurrence is linked to multiday cycles in diverse physiological signals. *Epilepsia*. 2023;64:1627–39.
14. Karoly PJ, Stirling RE, Freestone DR, Nurse ES, Maturana MI, Halliday AJ, et al. Multiday cycles of heart rate are

- associated with seizure likelihood: an observational cohort study. *EBioMedicine*. 2021;72:103619.
15. Vieluf S, El Atrache R, Hammond S, Touserkan FM, Loddenkemper T, Reinsberger C. Peripheral multimodal monitoring of ANS changes related to epilepsy. *Epilepsy Behav*. 2019;96:69–79.
 16. Proix T, Truccolo W, Leguia MG, Tchong TK, King-Stephens D, Rao VR, et al. Forecasting seizure risk in adults with focal epilepsy: a development and validation study. *Lancet Neurol*. 2021;20(2):127–35.
 17. Vieluf S, Amengual-Gual M, Zhang B, El Atrache R, Ufongene C, Jackson MC, et al. Twenty-four-hour patterns in electrodermal activity recordings of patients with and without epileptic seizures. *Epilepsia*. 2021;62(4):960–72.
 18. Vieluf S, El Atrache R, Cantley S, Jackson M, Clark J, Sheehan T, et al. Seizure-related differences in biosignal 24-h modulation patterns. *Sci Rep*. 2022;12(1):15070.
 19. Vieluf S, Cantley S, Krishnan V, Loddenkemper T. Ultradian rhythms in accelerometric and autonomic data vary based on seizure occurrence in paediatric epilepsy patients. *Brain Commun*. 2024;6(2):fcae034.
 20. Vieluf S, Cantley S, Jackson M, Zhang B, Bosl WJ, Loddenkemper T. Development of a multivariable seizure likelihood assessment based on clinical information and short autonomic activity recordings for children with epilepsy. *Pediatr Neurol*. 2023;148:118–27.
 21. Böttcher S, Vieluf S, Bruno E, Joseph B, Epitashvili N, Biondi A, et al. Data quality evaluation in wearable monitoring. *Sci Rep*. 2022;12(1):21412.
 22. Chollet F. Keras. 2015. GitHub. <https://github.com/fchollet/keras>
 23. O'Malley T, Bursztein E, Long J, Chollet F, Jin H, Invernizzi L, et al. Keras Tuner. 2019. <https://github.com/keras-team/keras-tuner>
 24. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: machine learning in python. *J Mach Learn Res*. 2011;12:2825–30.
 25. Karoly PJ, Ung H, Grayden DB, Kuhlmann L, Leyde K, Cook MJ, et al. The circadian profile of epilepsy improves seizure forecasting. *Brain*. 2017;140(8):2169–82.
 26. Karoly PJ, Eden D, Nurse ES, Cook MJ, Taylor J, Dumanis S, et al. Cycles of self-reported seizure likelihood correspond to yield of diagnostic epilepsy monitoring. *Epilepsia*. 2021;62(2):416–25.
 27. Gleichgerricht E, Dumitru M, Hartmann DA, Munsell BC, Kuzniecky R, Bonilha L, et al. Seizure forecasting using machine learning models trained by seizure diaries. *Physiol Meas*. 2022;43(12):124003.
 28. Al-Bakri AF, Villamar MF, Haddix C, Bensalem-Owen M, Sunderam S. Noninvasive seizure prediction using autonomic measurements in patients with refractory epilepsy. In: 2018 40th annual international conference of the IEEE engineering in medicine and biology society (EMBC). IEEE, pp. 2422–5 2018.
 29. Vieluf S, Hasija T, Schreier PJ, El Atrache R, Hammond S, Touserkan FM, et al. Generalized tonic-clonic seizures are accompanied by changes of interrelations within the autonomic nervous system. *Epilepsy Behav*. 2021;124:108321.
 30. Cereghino JJ. Identification and treatment of acute repetitive seizures in children and adults. *Curr Treat Options Neurol*. 2007;9(4):249–55.
 31. Goldenholz D, Brinkmann BH, Westover MB. How accurate do self-reported seizures need to be for effective medication management in epilepsy? *Epilepsia*. 2024;65(7):e104–e112. doi:10.1111/epi.18019
 32. Payne DE, Dell KL, Karoly PJ, Kremen V, Gerla V, Kuhlmann L, et al. Identifying seizure risk factors: a comparison of sleep, weather, and temporal features using a Bayesian forecast. *Epilepsia*. 2021;62(2):371–82.
 33. Seth EA, Watterson J, Xie J, Arulsamy A, Md Yusof HH, Ngadimon IW, et al. Feasibility of cardiac-based seizure detection and prediction: a systematic review of non-invasive wearable sensor-based studies. *Epilepsia Open*. 2024;9(1):41–59.
 34. Miron G, Halimeh M, Jeppesen J, Loddenkemper T, Meisel C. Autonomic biosignals, seizure detection, and forecasting. *Epilepsia*. 2024;1–14. <https://doi.org/10.1111/epi.18034>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Vieluf S, Tomioka S, Zhang B, Krishnan V, Bosl WJ, Grinnell T, et al. Seizure monitoring by combined diary and wearable data: A multicenter, longitudinal, observational study. *Epilepsia*. 2025;66:4259–4271. <https://doi.org/10.1111/epi.18550>